

A123 ANR26650M1B LFP Cell Physical Modeling and Calibration Using PyBaMM

1. Introduction

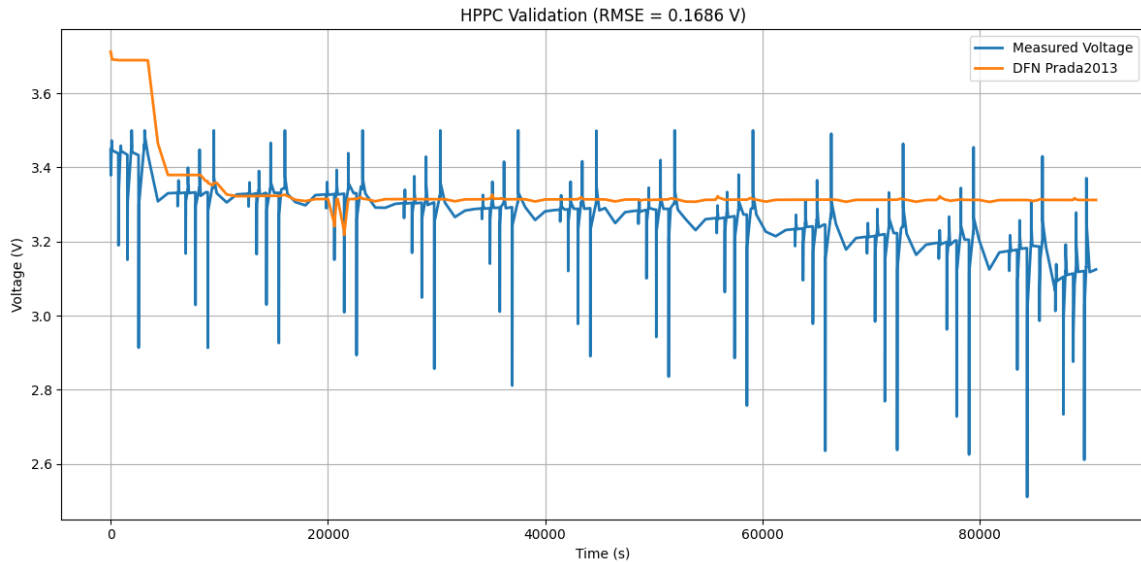
This study develops, calibrates, and validates a DFN electrochemical model for the A123 ANR26650M1B LFP/Graphite cell using PyBaMM. HPPC/discharge/charge datasets were used for calibration and validation.

2. Methodology

A DFN model with the Prada2013 parameter set was selected as the baseline. HPPC data were cleaned, synchronized, and applied as the model current profile. Calibration was performed sequentially to maintain physical interpretability.

3. Baseline DFN Simulation

The baseline DFN showed substantial voltage mismatch, capacity mismatch, and transient response errors. Baseline RMSE was approximately 0.17 V.



The above shown voltage simulation was run using the parameters of Prada2013 and current was scaled down by a factor of 10 to run the DFN code. However, there are multiple

fundamental issues in this simulation – evident from the voltage profile. In the subsequent steps, I have tried to counter those issues.

4. Calibration Strategy

Calibration was performed through OCP alignment, stoichiometry optimization, geometry calibration, diffusivity tuning, kinetic optimization, electrolyte correction, and MP-DFN development.

4.1. OCP Calibration

A literature-derived LFP cathode OCP was implemented together with a +0.10 V alignment shift to match measured voltage behaviour.

4.2. Stoichiometry Optimization

Differential evolution optimization produced $\theta_{n_init} = 0.905$ and $\theta_{p_init} = 0.5075$, significantly improving discharge voltage prediction.

4.3. Geometry Calibration

Negative electrode thickness, positive electrode thickness, and positive active material volume fraction were adjusted (with reference to the actual experimental values available) to improve capacity matching.

4.4. Dynamic Parameter Calibration

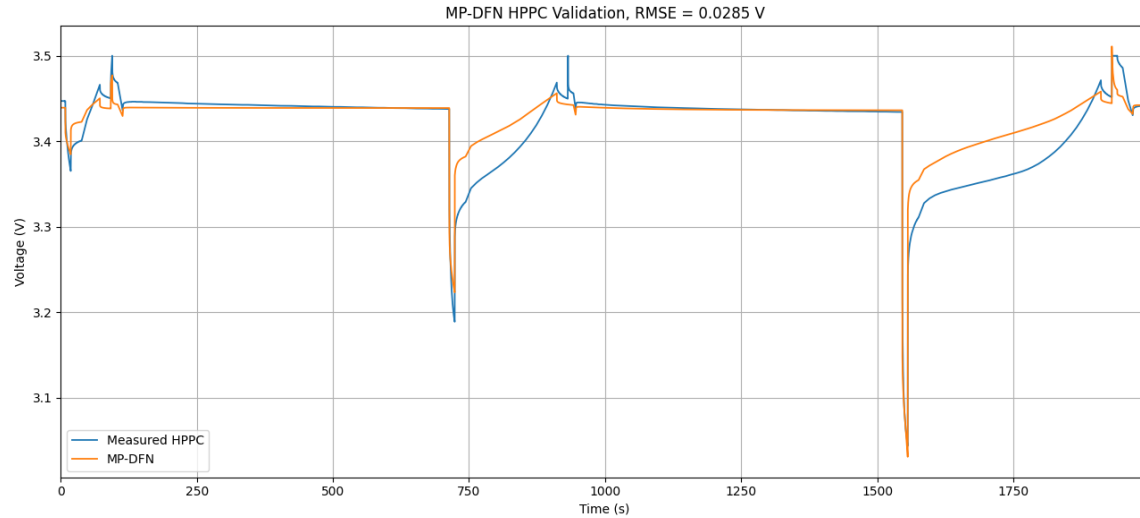
Positive diffusivity, exchange-current multipliers, and electrolyte conductivity were calibrated to reduce transient voltage errors.

4.5. MP-DFN Development

An MP-DFN model including particle-size distribution was implemented to better represent heterogeneous diffusion behaviour.

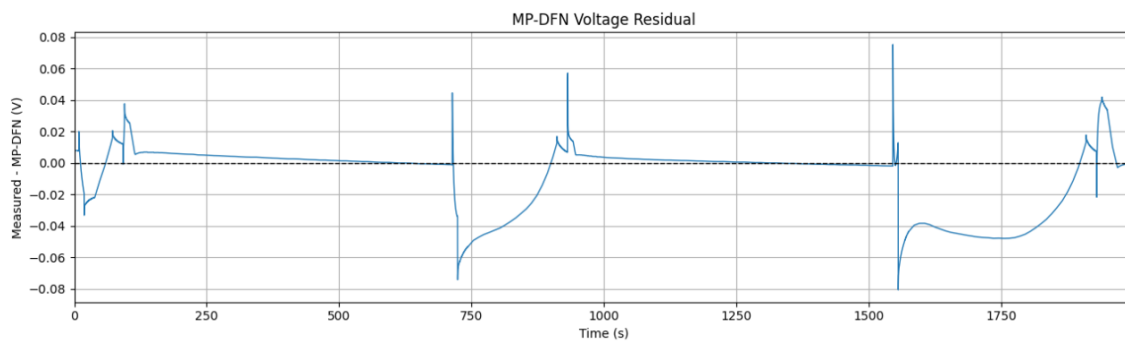
5. Conclusion

The final model reproduces major experimental behaviours while retaining physical meaning in calibrated parameters. Calibration reduced model error from approximately 0.19 V to approximately 26–30 mV.



After multiple optimization strategies, I was able to optimize the DFN model. But, the only challenge that need further improvement/investigation is hysteresis nature of OCV curve. The above diagram shows the result which capture the HPPC profile current to a good extent. This plot shows the simulation vs experiment data till $t = 2000s$. If code is utilized for the complete time interval, hysteresis issue will amplify.

Also, while plotting the charging curves of 1-4C data was analysed, an OCP offset with hysteresis was observed which confirmed my hypothesis of OCV hysteresis challenge. Although, due to submission deadline, that challenge is not fully addressed in this assignment.



For the simulated time window of HPPC, rest and discharge dynamics are captured well as compared to the charging pulse region. Similarly, when only 1C discharge behaviour was compared with the experimental data, it showed a good fit for optimized parameters. And, hysteresis behavior was observed while plotting the charging profiles. This was the reason I preferred to MP-DFN to somehow encounter the dynamics of phase change. In addition to that, I tried multiple approaches to cover the hysteresis effect but it impacted the overall accuracy of the model.

6. Results Summary

Baseline DFN RMSE ≈ 0.19 V

Calibrated DFN RMSE $\approx 26\text{--}30$ mV

MP-DFN RMSE ≈ 28 mV

RMSE was selected because it penalizes large voltage deviations more strongly than MAE and therefore provides a robust assessment of pulse and relaxation voltage prediction accuracies.

7. Future Work

Future work includes multi-rate charge calibration, explicit LFP hysteresis modelling, thermal coupling, PSD validation, and long-duration HPPC validation.

APPENDIX-1: Parameters

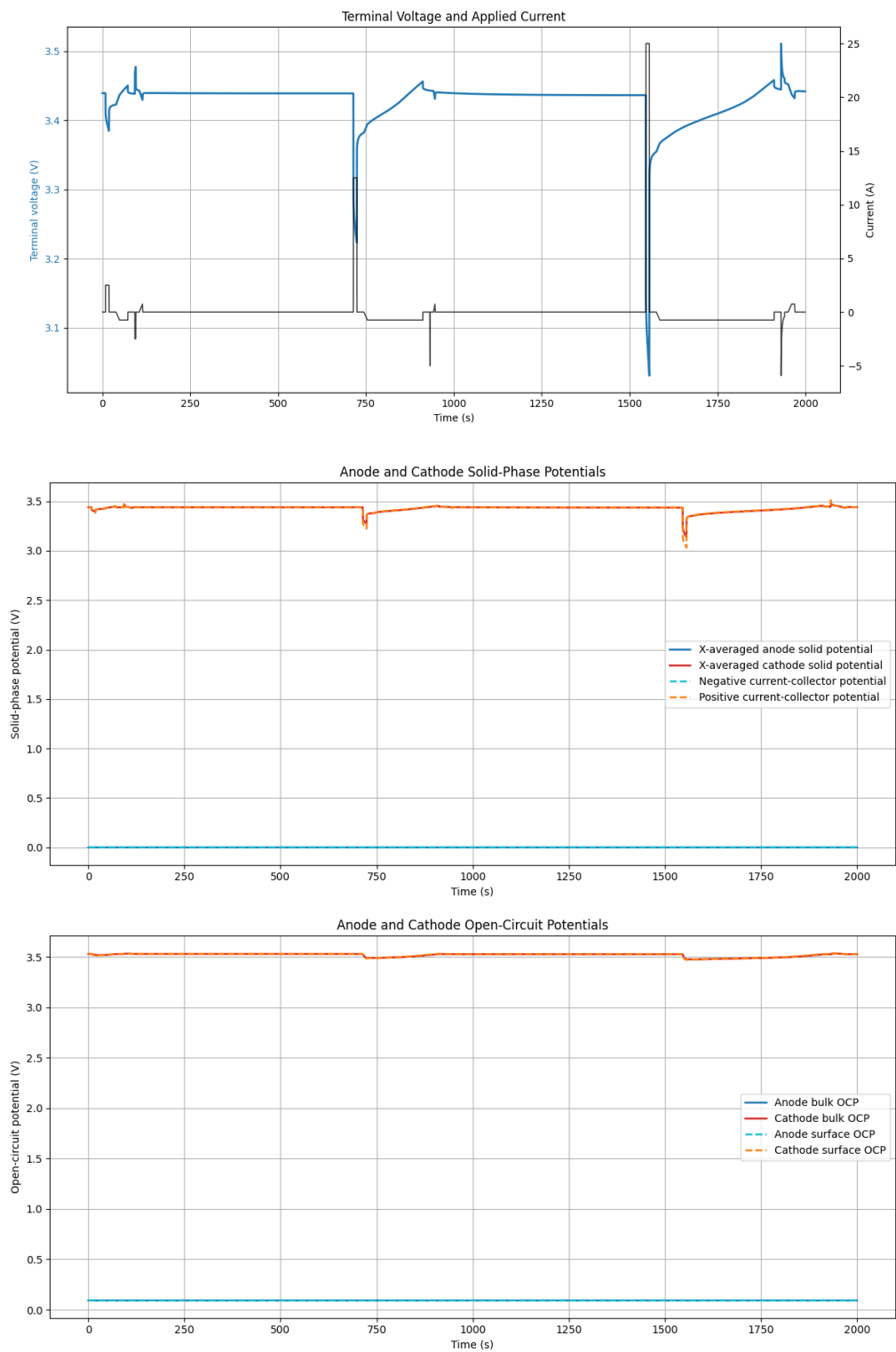
Optimization Progress Summary

Stage	Parameters Modified	Result
Baseline DFN	Default Prada2013	RMSE ~0.19 V
Stoichiometry Optimization	theta_n, theta_p	Major improvement
Geometry Calibration	Ln, Lp, volume fraction	Capacity matched
Dynamic Calibration	Diffusivity, kinetics, conductivity	Improved HPPC response
MP-DFN	Particle-size distribution	RMSE ~28 mV

Final Calibrated Parameters

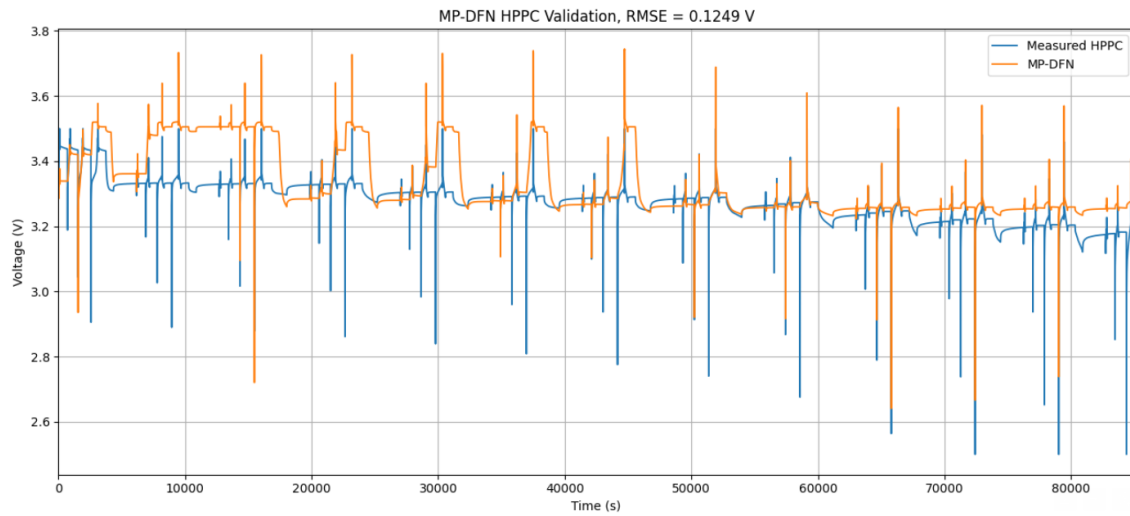
Parameter	Value	Remark
theta_n_init	0.905	Value obtained from parameter sweep
theta_p_init	0.5075	
Negative electrode thickness	129e-6 m	Considered actual values
Positive electrode thickness	171e-6 m	
Positive active material volume fraction	0.55	Value obtained from parameter sweep
Positive particle radius	2.5e-6 m	Considered actual value
Positive diffusivity	1.0e-13 m ² /s	Value obtained from parameter sweep
Positive kinetic multiplier	30	These parameters were tuned to fit the voltage profile
Negative kinetic multiplier	30	
Electrolyte conductivity multiplier	1.6	
Electrolyte diffusivity multiplier	1.0	
Contact resistance	0 Ω	
Temperature	296.62 K	
OCP shift	+0.10 V	

APPENDIX-2: Variable plots



APPENDIX-3: Full HPPC validation

To assess the generalizability of the identified parameter set, the optimized MP-DFN model was evaluated against the complete HPPC dataset. Unlike the calibration stage, which focused on a representative HPPC segment (2000-9000s) to refine transient response characteristics, this validation was performed over the entire test duration without further parameter adjustment. No additional empirical OCV shift was applied during this validation, allowing the predictive capability of the parameterized model to be assessed directly. The comparison demonstrates that the model captures the overall voltage trend and pulse-response behaviour across a wide SOC range, achieving a full-profile RMSE of 0.125 V. Remaining discrepancies are attributed to hysteresis effects, SOC-dependent polarization, and parameter variations outside the calibration region.



Reference

1. OCV plot references – [Paper 1](#) & [Paper 2](#)
2. Physical parameters like particle radii, electrodes thickness were taken from actual data available for LithiumWerks LFP cell.